

ZP-DOC-04

Use Case Model

Actors, use case diagrams, fully dressed use cases and activity flows

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Revision history

Version	Date	Author	Description of change
0.9	5 Oct 2026	Team Kestrel	First issue: 18 use cases (7 fully dressed), two use case diagrams, two activity diagrams; submitted for approval
0.9	5 Oct 2026	Team Kestrel	Review update: catalogue extended for FR-113 (O/L requirements) and FR-308 (list limit) added to ZP-DOC-03
1.0	5 Oct 2026	B.M.P Banneka	Approved by the product owner; baselined

Review and approval

This document takes effect when the approver signs below. Until then its status is as shown on the cover.

Role	Name	Signature	Date
Reviewer	Claude (AI coding partner), consistency and traceability review		
Approver	B.M.P Banneka, Product Owner	Approved in writing (project channel)	5 Oct 2026

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1 Introduction

1.1 Purpose

This document describes how each actor uses ZedPath to reach a goal. Where the user stories (ZP-DOC-02) say *what* a user needs and the SRS (ZP-DOC-03) says *what the system shall do*, the use cases describe the **interaction step by step**, including what happens when things go wrong. They are the basis for the screen flows (ZP-DOC-07), the API design (ZP-DOC-06) and the system tests (ZP-DOC-08).

1.2 Scope

Eighteen use cases covering Releases R1 and R2. Seven are written in Cockburn's *fully dressed* format because they carry the main risk or value; the rest are given in *brief* format.

1.3 Intended audience

Developers, testers, interface designers and reviewers.

1.4 Relationship to other documents

Table 1: Position of this document in the ZedPath document set

Document	Relationship
ZP-DOC-03 SRS	Parent. Each use case realises a group of functional requirements (listed per use case)
ZP-DOC-05 Data Design	Child. Nouns in the use cases name the entities
ZP-DOC-07 UI/UX Specification	Child. Main scenarios become screen flows
ZP-DOC-08 Test Plan	Child. Main scenarios and extensions become system test cases

1.5 Conventions

- Use case IDs are **UC-*nn***. Goal level follows Cockburn: **user goal** (sea level) unless stated; **subfunction** for included steps.
- In scenarios, numbered steps are the main success path; extensions are numbered by the step they branch from (for example *3a*).
- *The system* means ZedPath.

2 Actors

Table 2: Actors and their goals

Actor	Type	Goals
Student (P1, P2)	Primary, human	Find every reachable path, avoid costly mistakes, never miss a deadline
Family member (P3)	Secondary, human	Receive a clear summary in their language
Teacher (P4)	Primary, human (R2)	See which students need help
Verified senior (P5)	Primary, human (R2)	Help juniors in a trusted space
Curator (P6)	Primary, human	Keep every published fact correct and sourced
Scheduler	Primary, time	Run weekly Gazette scans and daily reminders
AI extractor (Gemini via AI Gateway)	Supporting, system	Propose structured facts from documents and photos; draft answers
AI checker (second model)	Supporting, system	Independently agree or disagree with each extracted fact
Push service	Supporting, system	Deliver web-push reminders to the student's browser

3 Use case diagrams

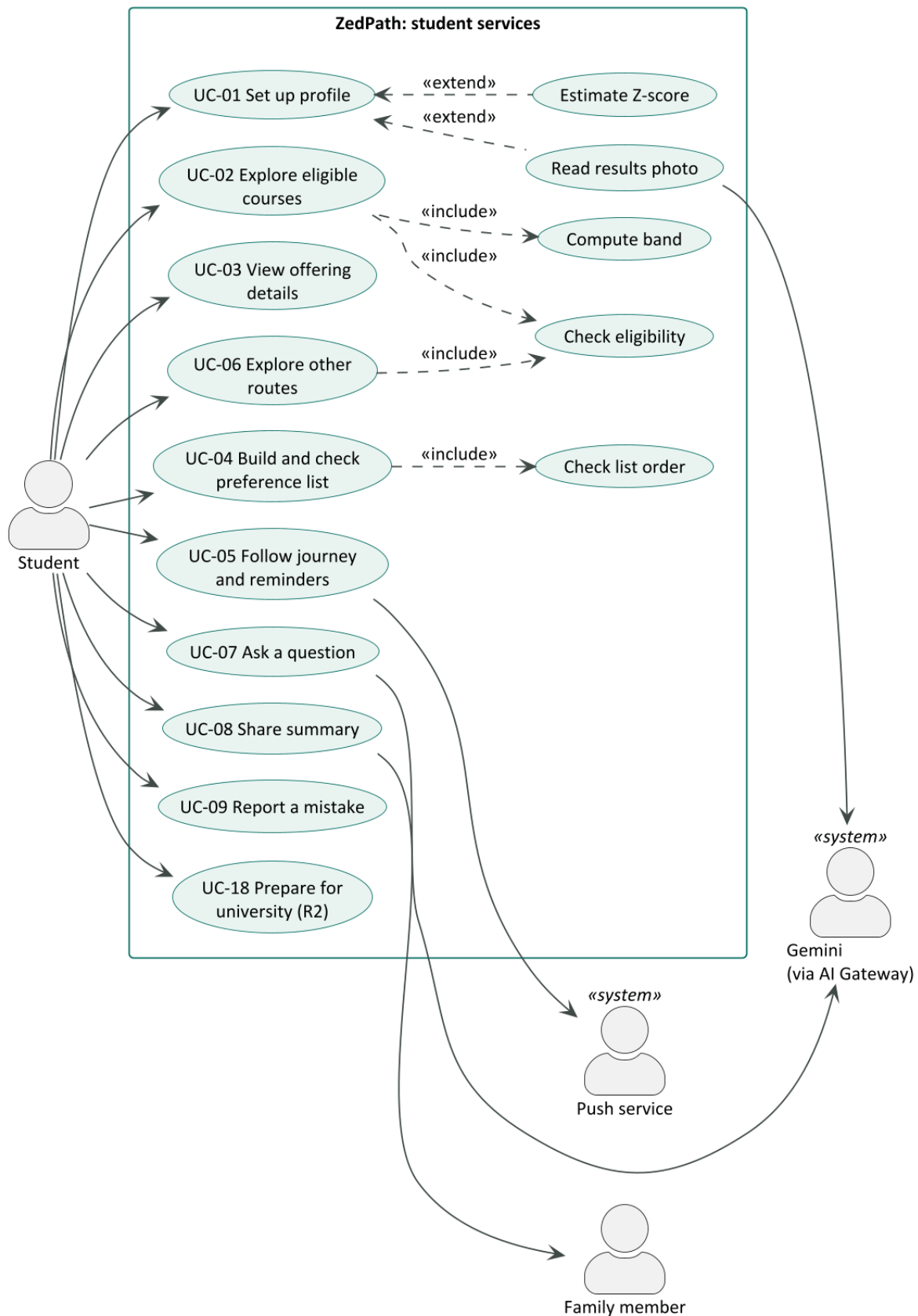


Figure 1: Student services: use cases of the student and the systems that support them

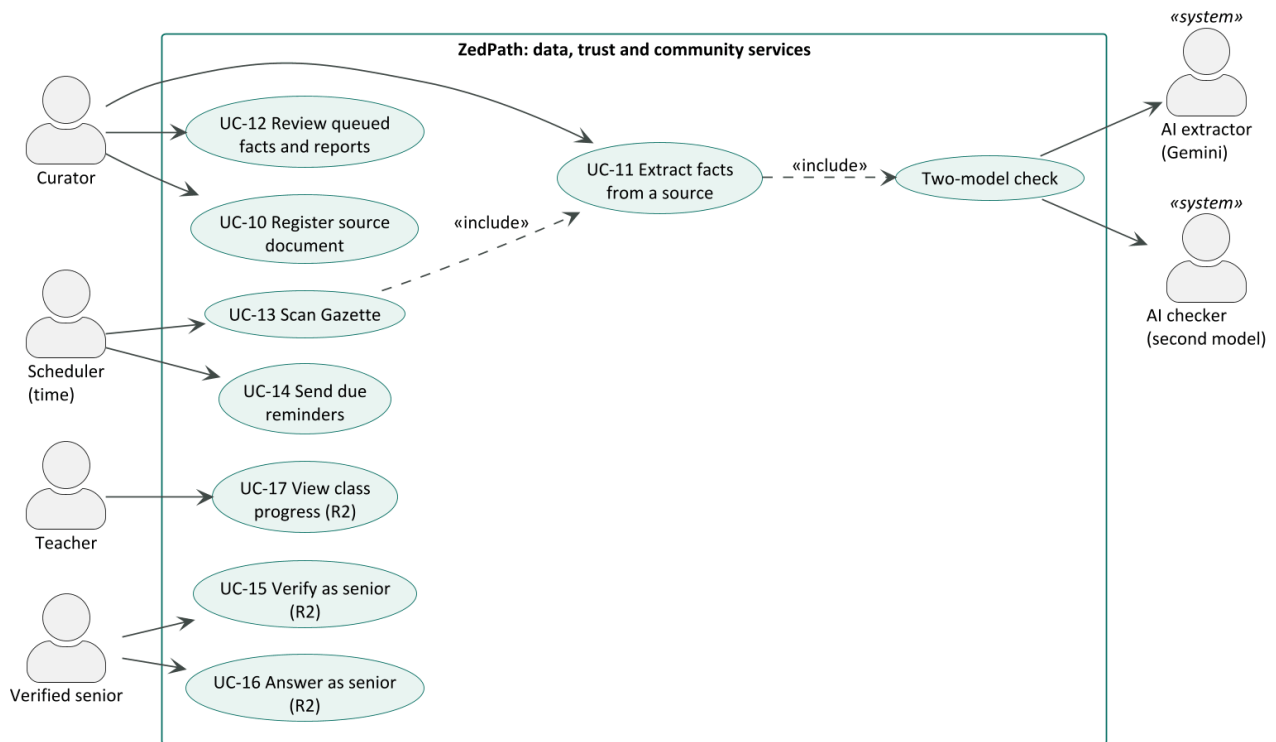


Figure 2: Data, trust and community services: use cases of the curator, scheduler, teacher and senior

4 Use case catalogue

Table 3: Use case catalogue with level, release and realised requirements

ID	Use case	Primary actor	Level	Release	Requirements
UC-01	Set up profile	Student	User goal	R1	FR-101 to FR-113
UC-02	Explore eligible courses	Student	User goal	R1	FR-201 to FR-207, FR-210 to FR-213
UC-03	View offering details	Student	User goal	R1	FR-208, FR-209, FR-214
UC-04	Build and check preference list	Student	User goal	R1	FR-301 to FR-308
UC-05	Follow journey and reminders	Student	User goal	R1	FR-401 to FR-408
UC-06	Explore other routes	Student	User goal	R1	FR-501 to FR-508
UC-07	Ask a question	Student	User goal	R1	FR-601 to FR-606, FR-913
UC-08	Share summary	Student	User goal	R1	FR-701, FR-702
UC-09	Report a mistake	Student	Subfunction	R1	FR-910, FR-912
UC-10	Register source document	Curator	User goal	R1	FR-901, FR-902
UC-11	Extract facts from a source	Curator, Scheduler	User goal	R1	FR-903 to FR-905
UC-12	Review queued facts and reports	Curator	User goal	R1	FR-906, FR-907, FR-911
UC-13	Scan Gazette	Scheduler	User goal	R1	FR-503, FR-504, FR-908, FR-909
UC-14	Send due reminders	Scheduler	User goal	R1	FR-405, FR-406
UC-15	Verify as senior	Verified senior	User goal	R2	FR-704
UC-16	Answer as senior	Verified senior	User goal	R2	FR-703, FR-705
UC-17	View class progress	Teacher	User goal	R2	FR-706
UC-18	Prepare for university	Student	User goal	R2	FR-801 to FR-804

5 Fully dressed use cases

5.1 UC-01 Set up profile

Scope	ZedPath (system)
Level	User goal
Primary actor	Student
Stakeholders and interests	Student: quick entry, no account, no mistakes. Product owner: no personal data on the server (NFR-030).
Preconditions	None. The student may be a first-time visitor.
Minimal guarantee	No profile data leaves the device unless the student uses photo reading (UC-01 extension 2a), and the photo is deleted.
Success guarantee	A valid profile is stored on the device and every results screen uses it.
Trigger	The student opens ZedPath without a stored profile, or chooses <i>Edit profile</i> .

Main success scenario

1. The student chooses a stream.
2. The student chooses the district they sat from.
3. The student enters their Z-score.
4. The student chooses three subjects and the grade for each.
5. The system validates the Z-score (BR-005) and the subject combination for the stream (BR-010).
6. The system stores the profile on the device and shows the student's results (UC-02).

Extensions

- *2a. The student has no results yet:* the student enters expected marks; the system shows an estimated Z-score range (FR-107) and continues at step 4 using the lower bound, marked *Estimate*.
- *2b. The student chooses "Read my results sheet":* the student photographs the results sheet; the system reads the fields (FR-109), highlights low-confidence fields (FR-110), deletes the photo (FR-111), and the student confirms or corrects each field; continue at step 5.
- *5a. The Z-score is invalid:* the system shows an inline message next to the field (FR-102); resume at step 3.
- *5b. The subject combination is not admissible for the stream:* the system names the subject that does not fit (FR-103); resume at step 4.
- **a. At any time the student switches language:* all text changes; entered data is kept (FR-112).

Frequency	Once per student, then occasional edits
Open issues	Subject statistics for the estimate need at least three years of published data (DE-2)

5.2 UC-02 Explore eligible courses

Scope	ZedPath (system)
Level	User goal
Primary actor	Student
Stakeholders and interests	Student: see every reachable course and an honest chance. Product owner: never imply certainty (FR-207).

Preconditions	A valid profile exists (UC-01). Cut-offs and rules for the intake year are published.
Minimal guarantee	Only courses whose rules the student meets are shown as eligible; no band is shown without data.
Success guarantee	The student sees eligible offerings grouped by band, and knows why any course is hidden.
Trigger	The student submits or changes their profile, or opens <i>Courses</i> .

Main success scenario

1. The system sends the profile to the server.
2. The server applies the general eligibility rules (BR-001 to BR-004) and each offering's specific rules (BR-020 to BR-024) (*include: Check eligibility*).
3. The server bands each eligible offering against the district's cut-off history (BR-030 to BR-036) (*include: Compute band*).
4. The system shows counts per band and the list for the default band, each item with its gap to the latest cut-off and its trend.
5. The student filters by band, university, field or distance (FR-210).
6. The student opens an offering (UC-03) or adds it to the list (UC-04).

Extensions

- *2a. Some offerings fail a rule:* the system shows "N courses hidden" with a link to the reasons (FR-204).
- *3a. An offering has no numeric cut-off for the district (all NQC or new):* it is shown as *Not enough data*.
- *3b. The offering is selected on all-island merit:* the all-island cut-off is used for every district (BR-035).
- *5a. The student marks an offering as a dream course:* the system shows the gap and up to three nearby open offerings (FR-211, FR-212).
- *1a. The network is unavailable:* the system shows the last results computed on the device, marked with the time computed.

Frequency	Several times per session during the application window
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5.3 UC-04 Build and check preference list

Scope	ZedPath (system)
Level	User goal
Primary actor	Student
Stakeholders and interests	Student: a list that cannot lock them out of a course they want more (E4).
Preconditions	UC-02 has run; at least one eligible offering exists.
Minimal guarantee	The list never contains an offering the student is not eligible for.
Success guarantee	The student has an ordered list with no unacknowledged ordering warnings, and has copied the uni-codes.
Trigger	The student taps <i>Add to my list</i> or opens <i>My list</i> .

Main success scenario

1. The student adds an eligible offering; the system appends it with its band and uni-code.
2. The student reorders the list by drag and drop or keyboard.

3. The system re-checks the order after every change (*include: Check list order*) and shows any warnings (BR-041, BR-042).
4. The student resolves or acknowledges each warning.
5. The student taps *Copy uni-codes*; the system copies them in order, one per line (FR-306).

Extensions

- *1a. The offering is not eligible*: the system refuses and shows the failed rule (FR-302).
- *3a. The student taps "Fix order for me"*: the system proposes an order (FR-305); the student confirms or cancels.
- *3b. The list exceeds the maximum number of preferences allowed (BR-043)*: the system prevents further additions and explains the limit.

The decision logic of this use case is shown below.

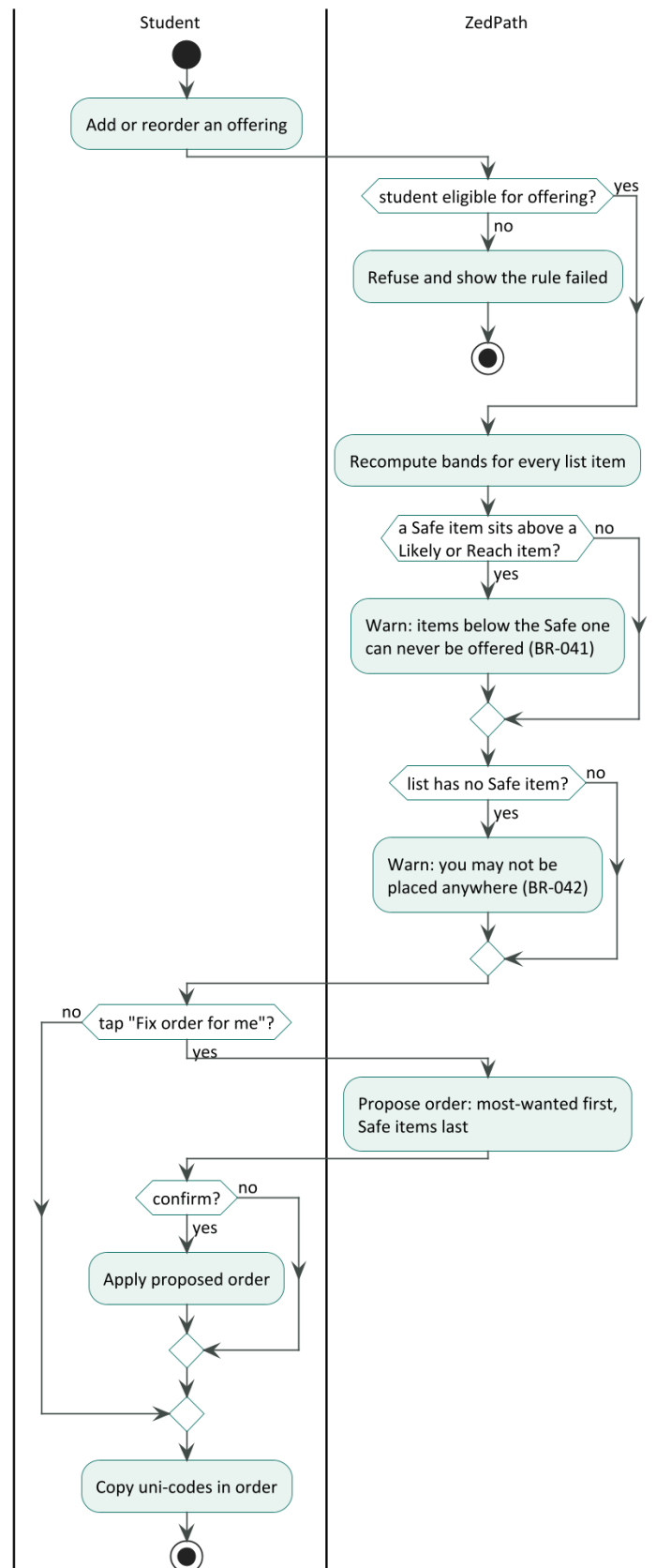


Figure 3: Activity flow of UC-04: adding, checking and fixing the preference list

5.4 UC-07 Ask a question

Scope	ZedPath (system)
Level	User goal
Primary actor	Student
Supporting actors	AI extractor (Gemini via AI Gateway)
Stakeholders and interests	Student: a correct answer in their language. Product owner: no unsupported or predictive answers (FR-604, FR-605); free-plan limits respected (NFR-034).
Preconditions	None; a profile improves answers.
Minimal guarantee	No answer is shown without a source; rate limits are enforced.
Success guarantee	The student receives an answer in their language with at least one source link.
Trigger	The student submits a question on <i>Ask</i> .

Main success scenario

1. The student types a question in Sinhala, Tamil or English.
2. The system passes the bot check and rate limit (FR-912, FR-913).
3. The system retrieves the stored source passages most relevant to the question.
4. The system asks the AI model to answer **only** from those passages, using the student's profile where relevant, in the question's language.
5. The system streams the answer with source chips linking to document and page.

Extensions

- *2a. Rate limit exceeded:* the system asks the student to wait and states when they can ask again.
- *3a. No relevant passage is found:* the system states it does not know and gives the official contact (FR-604).
- *4a. The question asks for a future cut-off or a guarantee:* the system declines and shows the historical pattern instead (FR-605).
- *4b. The AI service is unavailable:* the system says Ask is temporarily unavailable; all other features keep working (NFR-040).

5.5 UC-11 Extract facts from a source

Scope	ZedPath back office
Level	User goal
Primary actor	Curator (or Scheduler, from UC-13)
Supporting actors	AI extractor; AI checker
Stakeholders and interests	Students: correct rules. Curator: review only what needs review.
Preconditions	The source is registered and fingerprinted (UC-10).
Minimal guarantee	No fact is published without a source page and either two-model agreement or curator approval.
Success guarantee	Every extractable fact is either published or queued for review; the run is logged.
Trigger	The curator starts extraction, or a new Gazette issue arrives (UC-13).

Main success scenario

1. The system splits the source into sections with page numbers.
2. For each section, the AI extractor proposes structured facts with confidence and page.
3. The AI checker re-reads the page and agrees or disagrees with each fact (*include: Two-model check*).
4. Facts on which both agree with confidence at least 0.90 are published as new versions (FR-905).
5. All other facts enter the review queue (UC-12).
6. The system logs counts of published, queued and rejected facts.

Extensions

- *2a. A section cannot be read (scanned image without text):* the system uses vision extraction; if that fails, the section is queued for manual entry.
- *2b. The AI service fails or a limit is reached:* the step is retried with back-off (NFR-041); after three failures the run pauses and the curator is notified (FR-909).

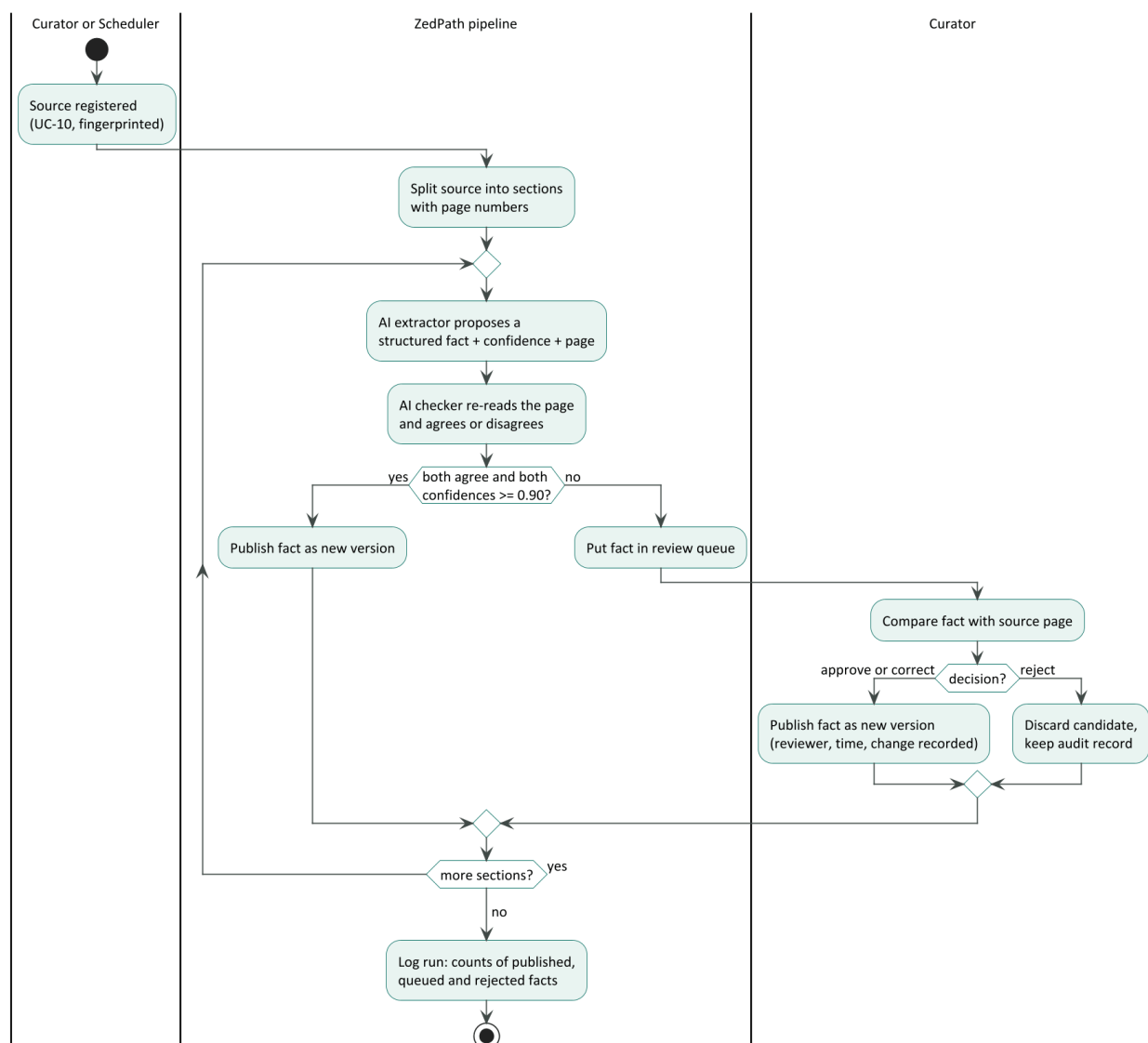


Figure 4: Activity flow of UC-11 and UC-12: extraction, two-model check and curator review

5.6 UC-12 Review queued facts and reports

Scope	ZedPath back office
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Level	User goal
Primary actor	Curator
Preconditions	The curator is signed in (NFR-032).
Success guarantee	Each reviewed item is approved, corrected or rejected, and the decision is recorded (FR-907).
Trigger	Items are waiting in the review queue or a mistake report arrives (UC-09).

Main success scenario

1. The curator opens the queue, ordered by impact (facts used by most students first).
2. The system shows the source page, the proposed fact and both models' verdicts side by side.
3. The curator approves, corrects or rejects the item.
4. The system publishes approved or corrected facts as new versions and keeps the previous version (FR-911).

Extensions

- *2a. The item is a mistake report:* the system shows the reported fact, the student's note and the source; the curator corrects the fact or closes the report with a reason.

5.7 UC-13 Scan Gazette

Scope	ZedPath back office
Level	User goal
Primary actor	Scheduler
Success guarantee	Every new Gazette issue is stored, its relevant notices extracted, and matching students alerted (FR-503).
Trigger	Weekly schedule (every Monday, 06:00 Sri Lanka time).

Main success scenario

1. The scheduler starts the weekly run.
2. The system fetches the list of new Gazette issues and stores each new file as a source (UC-10).
3. The system classifies notices and extracts job-examination details (UC-11).
4. Published examinations appear in the Alerts of students whose profile meets the education and age rules.

Extensions

- *2a. The Gazette website is unreachable:* the run retries; after three failures it is marked failed and shown to the curator (FR-909).

6 Brief use cases

Table 4: Brief use cases

ID	Use case	Brief description
UC-03	View offering details	The student opens an offering and sees its district cut-off history with their own Z-score, duration, medium, aptitude-test flag and first-year content, each with its source.
UC-05	Follow journey and reminders	The student sees dated stages with the next deadline first, each date confirmed or estimated, adds aptitude-test steps automatically, and may download a calendar file or turn on push reminders.
UC-06	Explore other routes	The student sees seven groups of non-UGC routes, each marked open or not open with the reason, and opens a route's requirements, cost and timing with sources.
UC-08	Share summary	The student generates a card in a chosen language, without name or Z-score unless added, and sends it through the phone's share sheet.
UC-09	Report a mistake	From any fact, the student taps <i>Report a mistake</i> , adds an optional note and passes a bot check; the report is queued for the curator.
UC-10	Register source document	The curator uploads an official document with its metadata; the system stores it unchanged, fingerprints it and rejects duplicates.
UC-14	Send due reminders	Daily, the scheduler finds deadlines due in 7 days or 1 day for subscribed students and sends web-push reminders; expired subscriptions are deleted.
UC-15	Verify as senior (R2)	A university student signs in with a university Google account on a recognised domain, or submits proof for curator review, and receives a verified badge.
UC-16	Answer as senior (R2)	A verified senior answers questions about their own course; answers show course, year and badge.
UC-17	View class progress (R2)	A teacher sees aggregate progress of students who shared with the class code, and names only of those who shared.
UC-18	Prepare for university (R2)	A selected student follows the university's registration checklist, checks Mahapola and bursary eligibility, plans the months before lectures and keeps a documents checklist.

Appendix A: Traceability: requirements to use cases

Every functional requirement in ZP-DOC-03 is realised by at least one use case, as listed in the use case catalogue above (column *Requirements*); this was checked by script against all 80 requirements.

Appendix B: Glossary

Table 5: Terms used in this document

Term	Meaning
Extension	An alternative path that branches from a step of the main success scenario
Fully dressed	Cockburn's complete use case format with stakeholders, guarantees, scenario and extensions
Include	A UML relationship: the base use case always performs the included use case
Extend	A UML relationship: the extending use case optionally adds behaviour to the base use case
Minimal guarantee	What the system promises even when the goal fails
Subfunction	A use case below user-goal level, performed as part of another

Appendix C: References

Table 6: Sources referred to in this document

Ref	Source
[1]	A. Cockburn, <i>Writing Effective Use Cases</i> , Addison-Wesley, 2001
[2]	Object Management Group, <i>Unified Modeling Language</i> , version 2.5.1, 2017
[3]	ZP-DOC-02 User Stories; ZP-DOC-03 Software Requirements Specification